

Basic Programs

BEGINNER LEVEL

Program 1: Variable Update and Display Write a program that declares an integer variable with initial value 10, updates it to 20, then adds 5 to it, then multiplies by 2, and prints the value after each update.

Program 2: Sum of Two Numbers Write a program that takes two numbers as input from the user and displays their sum.

Program 3: Simple Calculator Write a program that takes two numbers from the user and performs addition, subtraction, multiplication, division, and modulus on them, displaying all five results.

Program 4: Sum and Average of Three Numbers Write a program that takes three numbers as input and calculates their sum and average, then displays both results.

Program 5: Swap Two Numbers Using Third Variable Write a program that takes two numbers as input, swaps them using a third temporary variable, and displays values before and after swapping.

Program 6: Percentage Calculator Write a program that takes marks in 3 subjects (out of 100 each) as input, calculates total marks out of 300 and percentage, then displays both.

Program 7: Simple Interest Calculator Write a program that calculates simple interest using the formula: $SI = (P \times R \times T) / 100$, where P is principal, R is rate, and T is time.

Program 8: Celsius to Fahrenheit Converter Write a program that converts temperature from Celsius to Fahrenheit using the formula: $F = (C \times 9/5) + 32$

Program 9: Fahrenheit to Celsius Converter Write a program that converts temperature from Fahrenheit to Celsius using the formula: $C = (F - 32) \times 5/9$

Program 10: Area of Rectangle Write a program that calculates the area of a rectangle by taking length and width as input.

Program 11: Area of Circle Write a program that calculates the area of a circle by taking radius as input. Use formula: $\text{Area} = \pi \times r^2$ (use $\pi = 3.14159$)

Program 12: Area of Triangle Write a program that calculates the area of a triangle using the formula: $\text{Area} = (\text{base} \times \text{height}) / 2$

Program 13: Area and Perimeter of Square Write a program that takes the side of a square as input and calculates both its area and perimeter, then displays both results.

INTERMEDIATE LEVEL

Program 14: Area and Perimeter of Rectangle Write a program that takes length and width of a rectangle as input and calculates both its area and perimeter, then displays both results.

Program 15: Swap Two Numbers Without Third Variable Write a program that takes two numbers as input, swaps them without using any third variable (use arithmetic operations), and displays values before and after swapping.

Program 16: Days to Years, Weeks, Days Converter Write a program that takes total number of days as input and converts it into years, weeks, and remaining days, then displays all three values.

Program 17: Seconds to Hours:Minutes:Seconds Converter Write a program that converts total seconds into hours, minutes, and seconds format and displays the result.

Program 18: Average of Marks in 5 Subjects Write a program that takes marks obtained in 5 subjects as input, calculates the total marks and average marks, and displays both.

Program 19: Simple Interest with Total Amount Write a program that takes principal amount, rate of interest, and time period as input, calculates simple interest and total amount (Principal + SI), then displays both values.

Program 20: Profit or Loss Calculator Write a program that takes cost price and selling price as input, then determines and displays whether it's profit or loss and calculates the amount.

Program 21: Discount Price Calculator Write a program that calculates the final price after applying a discount percentage on the original price.

Program 22: Total Cost Calculator with Tax Write a program that calculates total cost of items: input price, quantity, and tax percentage, then calculate and display subtotal, tax amount, and final total amount.

Program 23: Electricity Bill Calculator Write a program that calculates electricity bill based on units consumed and rate per unit, then displays the total bill amount.

Program 24: BMI Calculator Write a program that calculates Body Mass Index using: $BMI = \text{weight}(\text{kg}) / (\text{height}(\text{m}))^2$